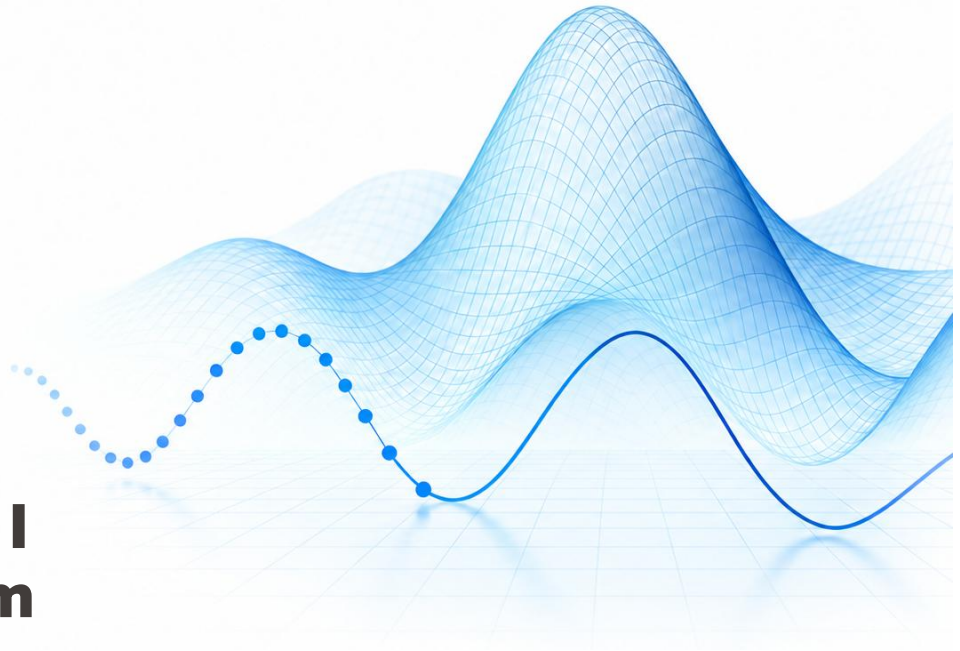


IE2106 – Engineering Math I
Week 9: Laplace Transform

Mohammad NIKOO
Nanyang Assistant Professor
Singapore NRF Fellow
EEE, NTU

We Learnt Last Week – Euler's Method

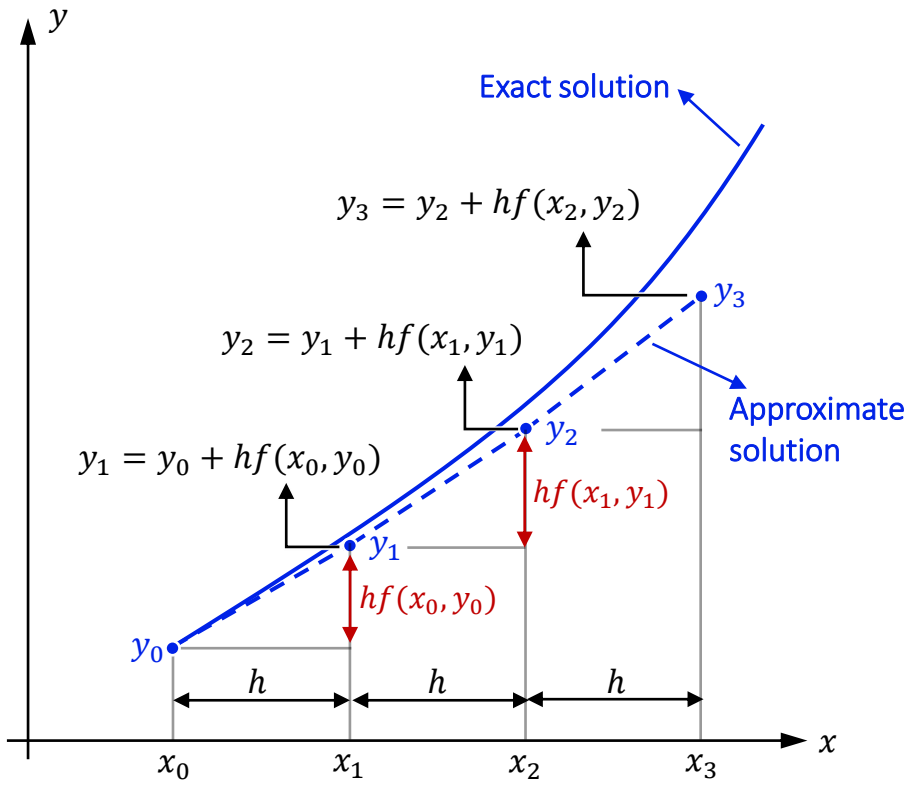
We consider a first-order ODE in a general form

$$y' = f(x, y), \qquad y(x_0) = y_0$$

f is a given function and y_0 is a given initial condition.

Euler's method considers a step size of h. It approximately computes $y_n \approx y(x_n)$ for $x_n = x_0 + nh$.

x_n	y_n (approximate solution at $x = x_n$)
x_0 (known)	y_0 (known)
$x_1 = x_0 + h$	$y_1 = y_0 + y_0' h = y_0 + f(x_0, y_0)h$
$x_2 = x_0 + 2h$	$y_2 = y_1 + y_1' h = y_1 + f(x_1, y_1)h$
...	...
$x_{n+1} = x_0 + (n+1)h$	$y_{n+1} = y_n + y_n' h = y_n + f(x_n, y_n)h$



First-, Second-, and Fourth-Order Approximation

To approximately solve $y' = f(x, y), y(x_0) = y_0$ with step size h

Euler's Method is a First-Order Approximation

It approximates y_{n+1} from y_n using a single-point slope approximation

$$a_n = f(x_n, y_n)h,$$

$$y_{n+1} = y_n + a_n$$

Improved Euler's Method is a Second-Order Approximation

It approximates y_{n+1} from y_n using a two-point slope approximation

$$a_n = f(x_n, y_n)h, \qquad b_n = f(x_{n+1}, y_n + a_n)h$$

$$y_{n+1} = y_n + \frac{1}{2}(a_n + b_n)$$

Fourth-order Runge–Kutta Method

It approximates y_{n+1} from y_n using a four-point slope approximation

$$a_n = f(x_n, y_n)h, \quad b_n = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}a_n\right)h, \quad c_n = f\left(x_n + \frac{1}{2}h, y_n + \frac{1}{2}b_n\right)h, \quad d_n = f(x_{n+1}, y_n + c_n)h$$

$$y_{n+1} = y_n + \frac{1}{6}(a_n + 2b_n + 2c_n + d_n)$$

Summary: Second-Order ODEs

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To approximately solve $y'' = f(x, y, y'), y(x_0) = y_0, y'(x_0) = y'_0$ with step size h

We define $y_1 = y$ and $y_2 = y'$. We have ① $y'_1 = y_2$, and ② $y'_2 = f(x, y_1, y_2)$ with initial conditions $y_{1,0} = y_0$ and $y_{2,0} = y'_0$.

Euler's Method is a First-Order Approximation

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a single-point slope approximation

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + a_{1,n}, \qquad y_{2,n+1} = y_{2,n} + a_{2,n}$$

Improved Euler's Method is a Second-Order Approximation

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a two-point slope approximation

$$a_{1,n} = y_{2,n}h, \qquad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h$$

$$b_{1,n} = (y_{2,n} + a_{2,n})h, \qquad b_{2,n} = f(x_{n+1}, y_{1,n} + a_{1,n}, y_{2,n} + a_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{2}(a_{1,n} + b_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{2}(a_{2,n} + b_{2,n})$$

Fourth-order Runge–Kutta Method

It approximates $y_{1,n+1}$ and $y_{2,n+1}$ using a four-point slope approximation

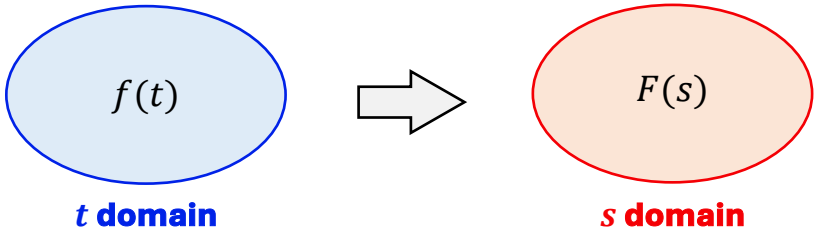
$$a_{1,n} = y_{2,n}h, \quad a_{2,n} = f(x_n, y_{1,n}, y_{2,n})h, \quad b_{1,n} = \left(y_{2,n} + \frac{1}{2}a_{2,n}\right)h, \quad b_{2,n} = f\left(x_n + \frac{1}{2}h, y_{1,n} + \frac{1}{2}a_{1,n}, y_{2,n} + \frac{1}{2}a_{2,n}\right)h$$

$$c_{1,n} = \left(y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \quad c_{2,n} = f\left(x_n + \frac{1}{2}h, y_{1,n} + \frac{1}{2}b_{1,n}, y_{2,n} + \frac{1}{2}b_{2,n}\right)h, \quad d_{1,n} = (y_{2,n} + c_{2,n})h, \quad d_{2,n} = f(x_{n+1}, y_{1,n} + c_{1,n}, y_{2,n} + c_{2,n})h$$

$$y_{1,n+1} = y_{1,n} + \frac{1}{6}(a_{1,n} + 2b_{1,n} + 2c_{1,n} + d_{1,n}), \quad y_{2,n+1} = y_{2,n} + \frac{1}{6}(a_{2,n} + 2b_{2,n} + 2c_{2,n} + d_{2,n})$$

Today's Objectives

Our main objective today is to learn Laplace Transform



- **Introduction to Step function, Delta (impulse) function, and Convolution operator**
- **Definition of Laplace Transform**
- **Properties of Laplace Transform**
- **Next week we will learn inverse Laplace Transform and applications of it.**

INTRODUCTION TO

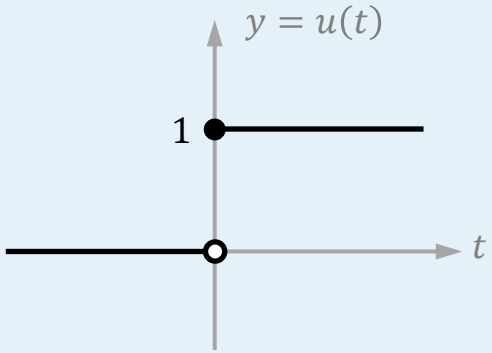
STEP FUNCTION, DELTA FUNCTION, AND CONVOLUTION

Step function and Delta function

Step Function

The step function is defined as

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$



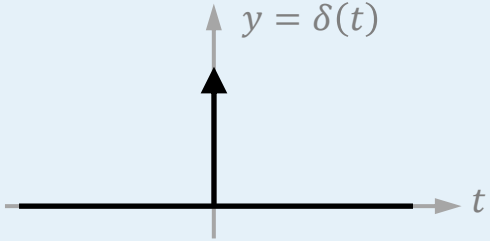
Delta Function

Delta function is the derivative of the step function

$$\delta(t) = \frac{\mathrm{d}}{\mathrm{d}t}u(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

Derivative is the reverse of integration, so we have:

$$\int_{-\infty}^{\infty} \delta(t)\,\mathrm{d}t = u(t)\Big|_{-\infty}^{+\infty} = u(+\infty) - u(-\infty) = 1 - 0 = 1$$

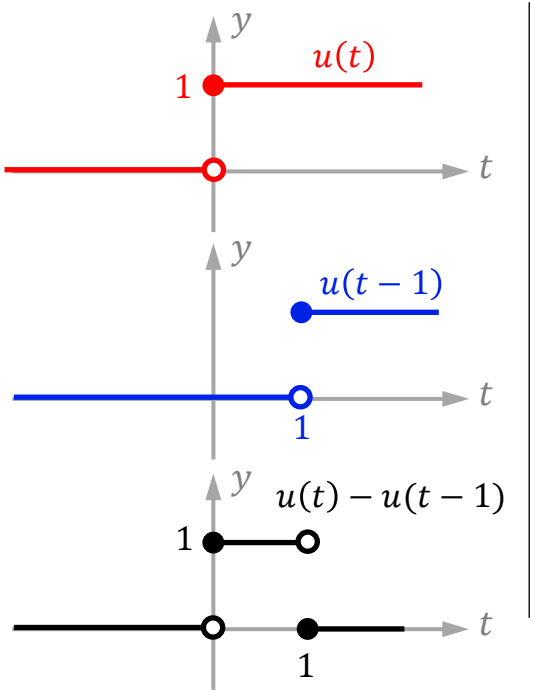


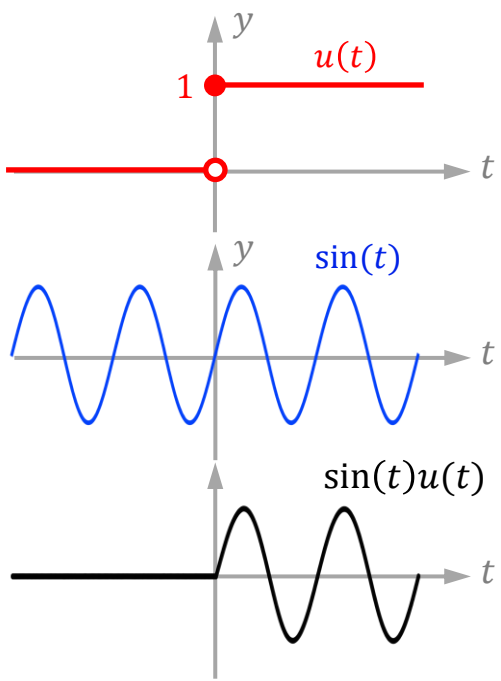
$$\boxed{\int_{-\infty}^{\infty} \delta(t)\,\mathrm{d}t = 1}$$

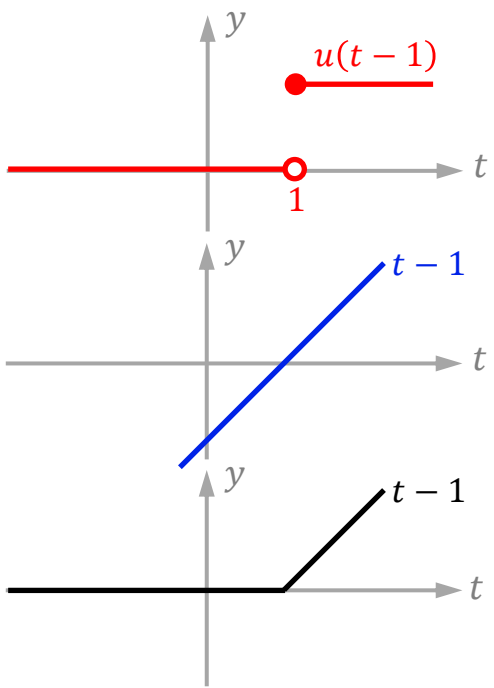
Example

Plot the following functions:

(a) $f(t) = u(t) - u(t-1)$ (b) $g(t) = \sin(t)u(t)$ (c) $h(t) = (t-1)u(t-1)$

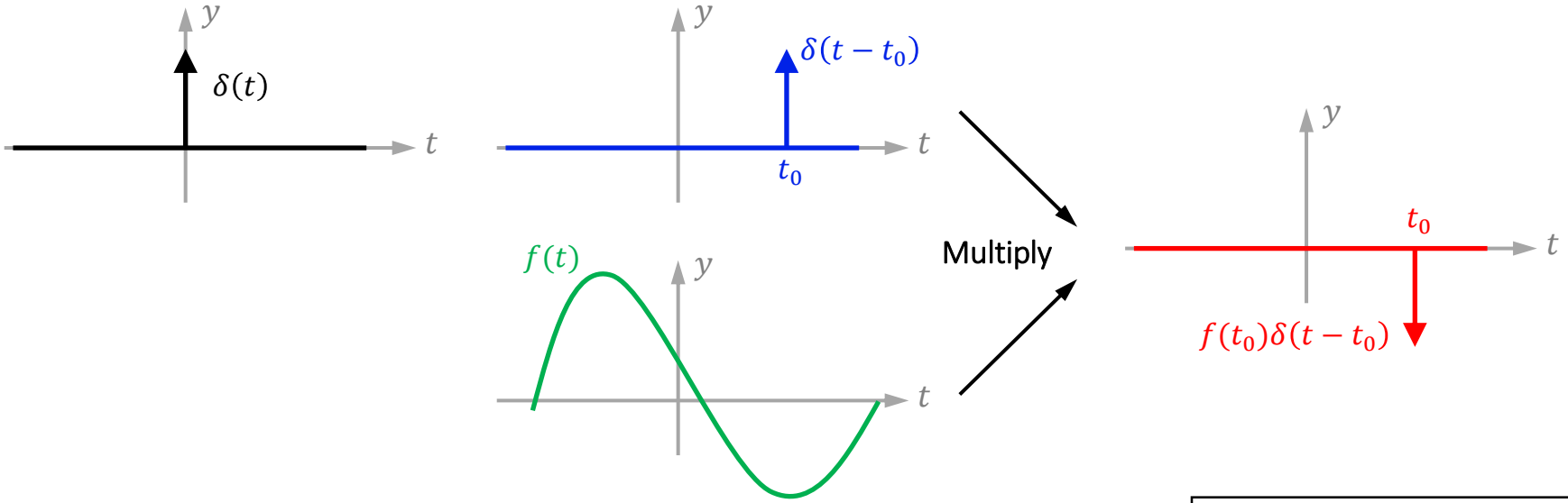






Example

Find $\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\,dt$



$$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\,dt = \int_{-\infty}^{\infty} f(t_0)\delta(t-t_0)\,dt = f(t_0)\underbrace{\int_{-\infty}^{\infty}\delta(t-t_0)\,dt}_{1} = f(t_0)$$

$$\boxed{\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\,dt = f(t_0)}$$

Convolution: $x(t) * h(t)$

Assuming two functions of the same variable, $x(t)$ and $h(t)$, we define convolution as

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

What does convolution do? For $y(t) = x(t) * h(t)$, let's consider $y(1)$ to see how this operator works:

$$y(1) = \int_{-\infty}^{\infty} x(\tau)h(1-\tau)\mathrm{d}\tau$$

So this is the way it works:

① Flip h (because of $-\tau$)

② Shift it (because of 1 in $1-\tau$)

③ Multiply it by $x(\tau)$

④ Integrate

Example

Considering $y(t) = \left(e^{-t}u(t)\right) * u(t)$, calculate $y(1)$

Using the definition of convolution, we write

$$y(t) = \underbrace{\left(e^{-t}u(t)\right)}_{x(t)} * \underbrace{u(t)}_{h(t)} = \int_{-\infty}^{\infty} \left(e^{-\tau}u(\tau)\right)u(t-\tau)\mathrm{d}\tau$$

To calculate $y(1)$ we have:

$$y(1) = \int_{-\infty}^{\infty} e^{-\tau}u(\tau)u(1-\tau)\mathrm{d}\tau$$

$u(\tau)u(1-\tau) = \begin{cases} 1, & 0 \le \tau \le 1 \\ 0, & \text{o. w.} \end{cases}$, so we write:

$$y(1) = \int_{0}^{1} e^{-\tau}\mathrm{d}\tau = 1 - e^{-1} = 0.632$$

Example

Compute $x(t) * \delta(t)$

Using the definition, we write:

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau$$

$x(\tau)\delta(t-\tau)$ is only non-zero for $\tau = t$. So we can write:

$$x(\tau)\delta(t-\tau) = x(t)\delta(t-\tau)$$

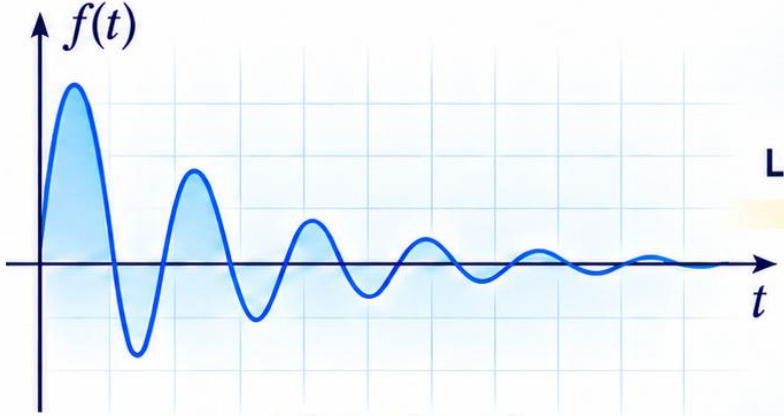
Therefore

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau = \int_{-\infty}^{\infty} x(t)\delta(t-\tau)\,\mathrm{d}\tau = x(t)\underbrace{\int_{-\infty}^{\infty} \delta(t-\tau)\,\mathrm{d}\tau}_{1} = x(t)$$

$\delta(t)$ is the identity element of convolution!

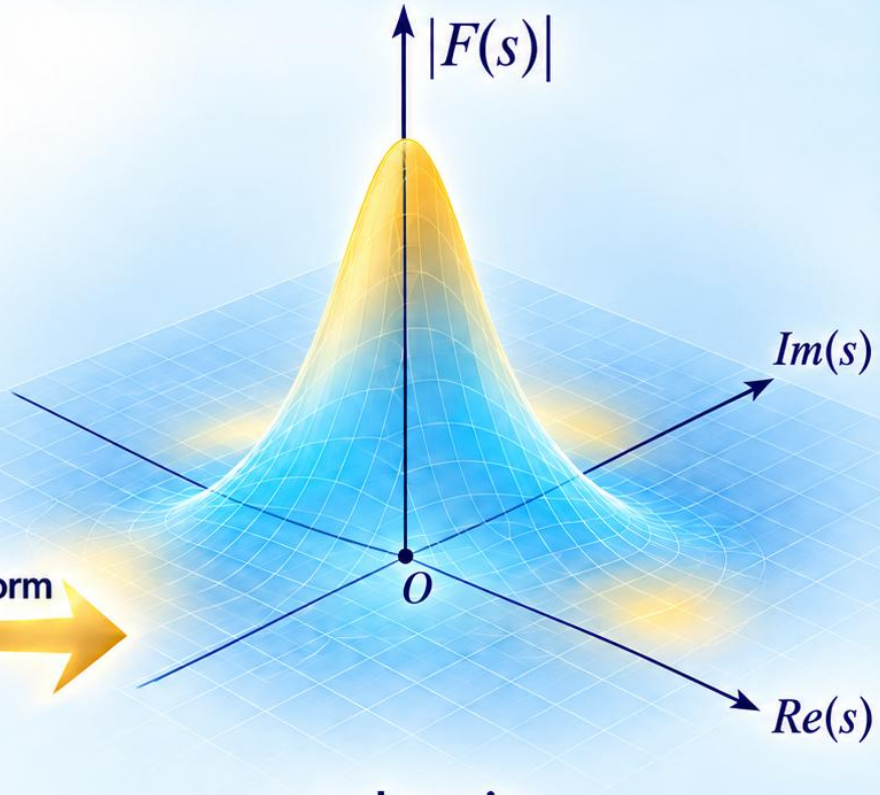
LAPLACE TRANSFORM

$$F(s) = \int_0^{\infty} f(t)e^{-st}\,dt$$



Time domain

Laplace transform



s-domain

$s = \sigma + j\omega$

From Function to Transformation

FUNCTION

Number x ⇨ Function ⇨ Number $f(x)$

Acts on numbers
The outputs are numbers too.

Example

$$f(x) = x^2$$

$$f(2) = 2^2 = 4$$

OPERATOR

Function $f(t)$ ⇨ Operator ⇨ Function $g(t)$

Acts on a function of t
The output is also a function of t

Example

$$\mathcal{D} = \frac{\mathrm{d}}{\mathrm{d}t}$$

$$\mathcal{D}(t^2) = 2t$$

TRANSFORM

Function $f(t)$ ⇨ Transform ⇨ Function $F(s)$

Acts on a function of t
The output is a function of s

Example

$$F(s) = \mathcal{L}(f(t)) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$$

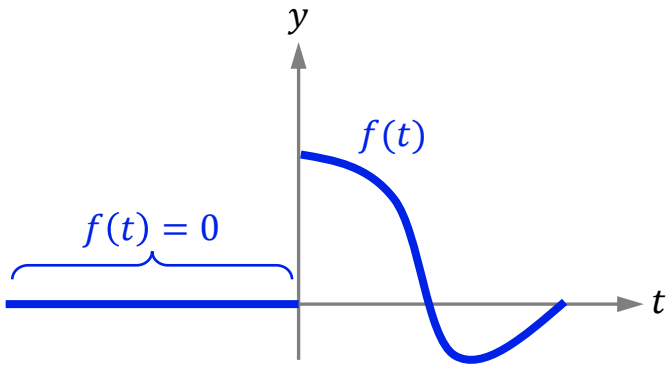
Laplace transform!

Laplace Transform

Laplace transform is a transformation from $\boldsymbol{t}$ domain to $\boldsymbol{s}$ domain.

$$F(s) = \mathcal{L}(f(t)) = \int_0^{\infty} f(t)e^{-st}\,\mathrm{d}t$$

- In this course, we work with single-sided definition of the Laplace transform, meaning that we start the integration from $t = 0$.
- Since we work with single-sided definition of the Laplace transform, we also assume that the function $f(t)$ is zero for negative time.



- Usually, t represents time and is a real variable, whereas s is a complex variable ($s = \sigma + j\omega$).

Example

Use the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$ to compute the Laplace transform of the following functions.

(a) $f(t) = u(t)$ (b) $g(t) = e^{-t}u(t)$ (c) $h(t) = \delta(t)$

(a) Based on the definition we have

$$F(s) = \mathcal{L}(u(t)) = \int_0^\infty u(t)e^{-st}\,\mathrm{d}t$$

We have $u(t) = 1$ for $t > 0$, so we write

$$F(s) = \int_0^\infty e^{-st}\,\mathrm{d}t = -\frac{1}{s}e^{-st}\Big|_0^\infty = -\frac{1}{s}(0-1) = \frac{1}{s}$$

So we have

$$\boxed{F(s) = \frac{1}{s}}$$

Example

Use the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$ to compute the Laplace transform of the following functions.

(a) $f(t) = u(t)$ (b) $g(t) = e^{-t}u(t)$ (c) $h(t) = \delta(t)$

(b) Based on the definition we have

$$G(s) = \mathcal{L}(e^{-t}u(t)) = \int_0^\infty e^{-t}u(t)e^{-st}\,\mathrm{d}t = \int_0^\infty e^{-t}e^{-st}\,\mathrm{d}t$$

We write

$$G(s) = \int_0^\infty e^{-(s+1)t}\,\mathrm{d}t = -\frac{1}{s+1}e^{-(s+1)t}\Big|_0^\infty$$

So we have

$$\boxed{G(s) = \frac{1}{s+1}}$$

Example

Use the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$ to compute the Laplace transform of the following functions.

(a) $f(t) = u(t)$ (b) $g(t) = e^{-t}u(t)$ (c) $h(t) = \delta(t)$

(c) Based on the definition we have

$$H(s) = \mathcal{L}(\delta(t)) = \int_0^\infty \delta(t)e^{-st}\,\mathrm{d}t$$

We use the property of delta function

$$\int_{-\infty}^{\infty} \delta(t)f(t)\,\mathrm{d}t = f(0)$$

Therefore, we have:

$$H(s) = \int_0^\infty \delta(t)e^{-st}\,\mathrm{d}t = e^{-s\cdot 0} = 1$$

$$\boxed{H(s) = 1}$$

Class Exercise

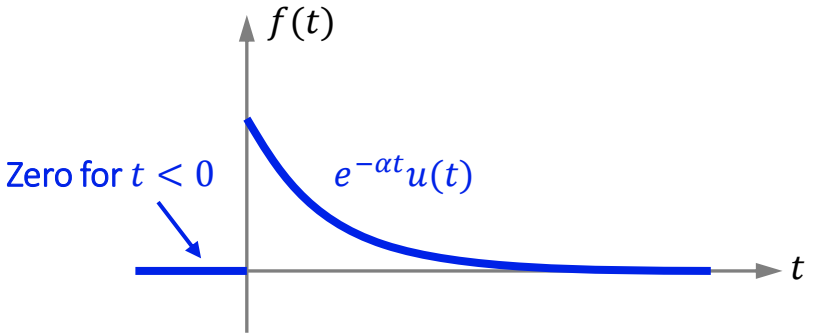
Plot $f(t) = e^{-\alpha t}u(t)$ assuming $\alpha > 0$ and compute its Laplace transform using the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$

Class Exercise

Plot $f(t) = e^{-\alpha t}u(t)$ assuming $\alpha > 0$ and compute its Laplace transform using the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$

Based on the definition we have

$$F(s) = \mathcal{L}(e^{-\alpha t}u(t)) = \int_0^\infty e^{-\alpha t}u(t)e^{-st}\,\mathrm{d}t = \int_0^\infty e^{-\alpha t}e^{-st}\,\mathrm{d}t$$



We write

$$F(s) = \int_0^\infty e^{-(s+\alpha)t}\,\mathrm{d}t = -\frac{1}{s+\alpha}e^{-(s+\alpha)t}\Big|_0^\infty$$

So we have

$$\boxed{F(s) = \frac{1}{s+\alpha}}$$

Class Exercise

Plot $f(t) = u(t-2)$ and compute its Laplace transform using the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$

Class Exercise

Plot $f(t) = u(t-2)$ and compute its Laplace transform using the definition $F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$

Based on the definition we have

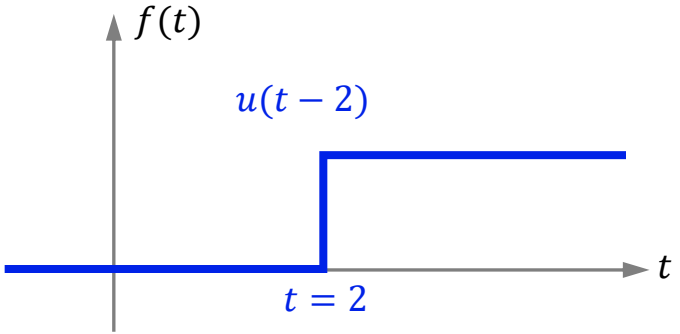
$$F(s) = \mathcal{L}(u(t-2)) = \int_0^\infty u(t-2)e^{-st}\,\mathrm{d}t = \int_2^\infty e^{-st}\,\mathrm{d}t$$

We write

$$F(s) = \int_2^\infty e^{-st}\,\mathrm{d}t = -\frac{1}{s}e^{-st}\Big|_2^\infty = 0 - \left(-\frac{1}{s}e^{-2s}\right)$$

So we have

$$\boxed{F(s) = \frac{1}{s}e^{-2s}}$$



Integration trick (for $t_0 > 0$):

$$\int_0^\infty u(t-t_0)f(t)dt = \int_{t_0}^\infty f(t)dt$$

PROPERTIES OF LAPLACE TRANSFORM

Example

Assuming $\mathcal{L}\big(f(t)\big) = F(s)$, compute $\mathcal{L}(f'(t))$

We start from definition of the Laplace transform: $\mathcal{L}\big(f'(t)\big) = \int_0^\infty f'(t)e^{-st}\,\mathrm{d}t$

Recap from calculus: Integration by parts

$$\int f'(t)g(t)\,\mathrm{d}t = f(t)g(t) - \int f(t)g'(t)\,\mathrm{d}t$$

So in this case we have ($g(t) = e^{-st}$):

$$\int f'(t)e^{-st}\mathrm{d}t = f(t)e^{-st} - \int f(t)(-se^{-st})\mathrm{d}t = f(t)e^{-st} + s\int f(t)e^{-st}\mathrm{d}t$$

We can integrate from 0 to ∞

$$\int_0^\infty f'(t)e^{-st}\,\mathrm{d}t = \underbrace{f(t)e^{-st}\Big|_0^\infty}_{=0-f(0)} + s\underbrace{\int_0^\infty f(t)e^{-st}\,\mathrm{d}t}_{=F(s)} \quad \Longrightarrow \quad \int_0^\infty f'(t)e^{-st}\,\mathrm{d}t = sF(s) - f(0)$$

$$\boxed{\mathcal{L}(f'(t)) = s\mathcal{L}(f(t)) - f(0)}$$

Example

Assuming $\mathcal{L}(f(t)u(t)) = F(s)$, compute $\mathcal{L}(f(t-3)u(t-3))$

We start from definition of the Laplace transform:

$$\mathcal{L}(f(t-3)u(t-3)) = \int_0^\infty f(t-3)u(t-3)e^{-st}\,\mathrm{d}t$$

We define a new variable: $\tau = t - 3$

$$\int_0^\infty f(t-3)u(t-3)e^{-st}\,\mathrm{d}t = \int_{-3}^\infty f(\tau)u(\tau)e^{-s(\tau+3)}\,\mathrm{d}\tau = \int_0^\infty f(\tau)u(\tau)e^{-s\tau}e^{-3s}\mathrm{d}\tau = e^{-3s}\underbrace{\int_0^\infty f(\tau)u(\tau)e^{-s\tau}\mathrm{d}\tau}_{=F(s)}$$

So we obtain: $\boxed{\mathcal{L}(f(t-3)u(t-3)) = e^{-3s}F(s)}$

More generally, we can write: $\boxed{\mathcal{L}(f(t-t_0)u(t-t_0)) = e^{-t_0 s}F(s)}$

Table of Laplace Transform

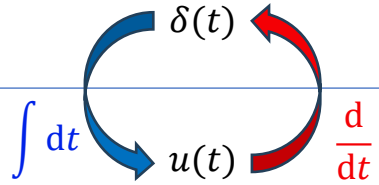
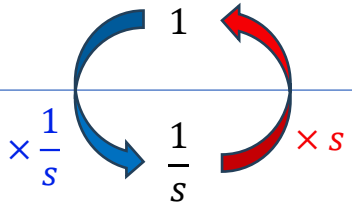
TIME DOMAIN	LAPLACE DOMAIN
	
$t\,u(t)$	$\frac{1}{s^2}$
$\frac{1}{2}t^2\,u(t)$	$\frac{1}{s^3}$
$\frac{1}{n!}t^n\,u(t)$	$\frac{1}{s^{n+1}}$

Table of Laplace Transform

TIME DOMAIN	LAPLACE TRANSFORM
$f(t)u(t)$	$F(s)$
$\int_0^t f(\tau)u(\tau)\,\mathrm{d}\tau$	$\frac{1}{s}F(s)$
$f'(t)u(t)$	$sF(s)-f(0)$
$f''(t)u(t)$	$s^2F(s)-sf(0)-f'(0)$
$f^{(n)}(t)u(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)-\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$

Table of Laplace Transform

TIME DOMAIN	LAPLACE TRANSFORM
$e^{-at}\,u(t)$	$\frac{1}{s+a}$
$t\,e^{-at}\,u(t)$	$\frac{1}{(s+a)^2}$
$t^2 e^{-at}\,u(t)$	$\frac{2!}{(s+a)^3}$
$t^n e^{-at}\,u(t)$	$\frac{n!}{(s+a)^{n+1}}$
$t^n f(t) u(t)$	$(-1)^n \frac{\mathrm{d}^n}{\mathrm{d}s^n} F(s)$

Table of Laplace Transform

TIME DOMAIN	LAPLACE TRANSFORM
$\sin(\omega t)\,u(t)$	$\dfrac{\omega}{s^2+\omega^2}$
$\cos(\omega t)\,u(t)$	$\dfrac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)\,u(t)$	$\dfrac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)\,u(t)$	$\dfrac{s+a}{(s+a)^2+\omega^2}$
$e^{-at}f(t)u(t)$	$F(s+a)$

Table of Laplace Transform

TIME DOMAIN	LAPLACE TRANSFORM
$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
$f(t - t_0)u(t - t_0), \quad t_0 > 0$	$e^{-st_0}F(s)$
$f(at)u(t), \quad a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
$f(t) * g(t)$	$F(s)G(s)$

Example

Consider $f(t) = \begin{cases} 0, & t < 0 \\ 1, & 0 \le t < 1 \\ 0, & 1 \le t \end{cases}$, (a) plot the function and write it based on the step function. (b) Compute $F(s)$.

(b) We can write $f(t)$ in form of step functions

$$f(t) = u(t) - u(t-1)$$

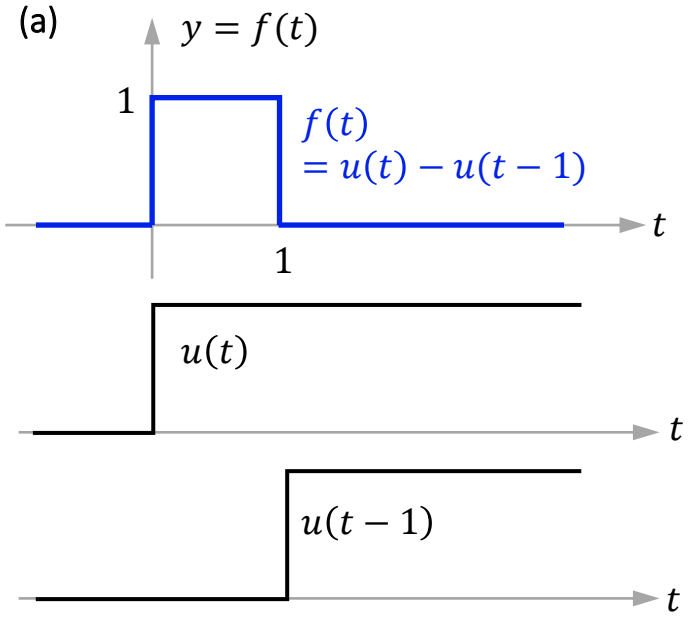
Using properties of Laplace transform we have

$$\mathcal{L}\big(u(t)\big) = \frac{1}{s}, \qquad \mathcal{L}\big(u(t-1)\big) = e^{-s}\mathcal{L}\big(u(t)\big) = \frac{e^{-s}}{s}$$

So we have

$$\boxed{F(s) = \frac{1 - e^{-s}}{s}}$$

(a)



Example

Compute Laplace transform of $f(t) = \sin(t)\cos(2t)\,u(t)$

There is no general rule for the Laplace transform of product of two functions. However, here we can transfer product to summation!

From trigonometry we know $\sin(a)\cos(b) = \frac{1}{2}(\sin(a+b) + \sin(a-b))$

$$\sin(t)\cos(2t) = \frac{1}{2}(\sin(3t) + \sin(-t)) = \frac{1}{2}(\sin(3t) - \sin(t))$$

From the Table of Laplace transform we know:

$$\mathcal{L}\big(\sin(3t)\,u(t)\big) = \frac{3}{s^2+9}, \qquad \mathcal{L}\big(\sin(t)\,u(t)\big) = \frac{1}{s^2+1}$$

So:

$$\boxed{\mathcal{L}\big(f(t)\big) = \frac{1}{2}\left(\frac{3}{s^2+9} - \frac{1}{s^2+1}\right)}$$

Class Exercise

Compute the Laplace transform of the following functions using the table of Laplace transforms.

(a) $f(t) = \left(3t^2 - 4t + 2\right)u(t)$ (b) $g(t) = \left(2e^{3t} + 5\sin(4t)\right)u(t)$ (c) $h(t) = \left(te^{1-2t}\right)u(t)$

Class Exercise

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(a)

$$f(t) = \left(3t^2 - 4t + 2\right)u(t) = 3t^2u(t) - 4tu(t) + 2u(t)$$

$$\mathcal{L}\left(t^2u(t)\right) = \frac{2}{s^3} \quad , \quad \mathcal{L}\left(tu(t)\right) = \frac{1}{s^2} \quad , \quad \mathcal{L}\left(u(t)\right) = \frac{1}{s}$$

$$F(s) = 3\left(\frac{2}{s^3}\right) - 4\left(\frac{1}{s^2}\right) + 2\left(\frac{1}{s}\right)$$

$$\boxed{F(s) = \frac{6}{s^3} - \frac{4}{s^2} + \frac{2}{s}}$$

Class Exercise

Compute the Laplace transform of the following functions using the table of Laplace transforms.

(a) $f(t) = \big(3t^2 - 4t + 2\big)u(t)$ (b) $g(t) = \big(2e^{3t} + 5\sin(4t)\big)u(t)$ (c) $h(t) = \big(te^{1-2t}\big)u(t)$

(b)

$$g(t) = \big(2e^{3t} + 5\sin(4t)\big)u(t) = 2e^{3t}u(t) + 5\sin(4t)\,u(t)$$

$$\mathcal{L}\big(e^{3t}u(t)\big) = \frac{1}{s-3} \quad , \quad \mathcal{L}\big(\sin(4t)\,u(t)\big) = \frac{4}{s^2+16}$$

$$G(s) = 2\left(\frac{1}{s-3}\right) + 5\left(\frac{4}{s^2+16}\right)$$

$$\boxed{G(s) = \frac{2}{s-3} + \frac{20}{s^2+16}}$$

Class Exercise

Compute the Laplace transform of the following functions using the table of Laplace transforms.

(a) $f(t) = \left(3t^2 - 4t + 2\right)u(t)$ (b) $g(t) = \left(2e^{3t} + 5\sin(4t)\right)u(t)$ (c) $h(t) = \left(te^{1-2t}\right)u(t)$

(c)

$$h(t) = \left(te^{1-2t}\right)u(t) = e \cdot \left(te^{-2t}\right)u(t)$$

$$\mathcal{L}\left(e^{-2t}\right) = \frac{1}{s+2} \quad , \quad \mathcal{L}\left(te^{-2t}\right) = \frac{1}{(s+2)^2}$$

$$H(s) = e\left(\frac{1}{(s+2)^2}\right)$$

$$\boxed{H(s) = \frac{e}{(s+2)^2}}$$

Class Exercise

Compute Laplace transform of $f(t) = \left(t^2 - 2t + 1\right)e^{-5t}u(t-1)$

Class Exercise

Compute Laplace transform of $f(t) = (t^2 - 2t + 1)e^{-5t}u(t-1)$

$$f(t) = (t-1)^2 e^{-5t} u(t-1)$$

$$f(t) = e^{-5}(t-1)^2 e^{-5(t-1)} u(t-1)$$

We can use the time-shift property. We define:

$$g(t) = e^{-5} t^2 e^{-5t} u(t)$$

Because $f(t) = g(t-1)$, we have $F(s) = e^{-s}G(s)$

So we can find $F(s)$ if we find $G(s)$

$$t^2 u(t) \xrightarrow{\mathcal{L}} \frac{2}{s^3} \qquad t^2 e^{-5t} u(t) \xrightarrow{\mathcal{L}} \frac{2}{(s+5)^3} \qquad e^{-5} t^2 e^{-5t} u(t) \xrightarrow{\mathcal{L}} \frac{2e^{-5}}{(s+5)^3}$$

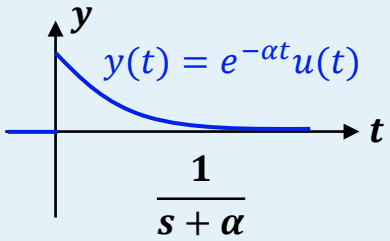
So $G(s) = \frac{2e^{-5}}{(s+5)^3}$ and we obtain $F(s) = e^{-s}G(s)$:

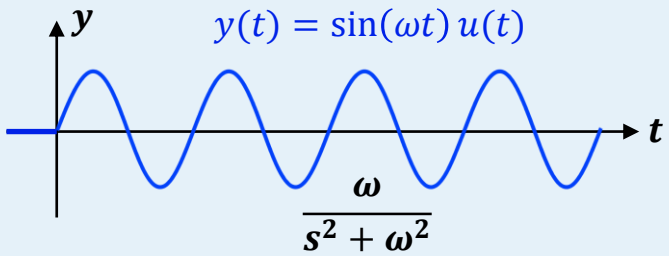
$$\boxed{F(s) = \frac{2e^{-5}e^{-s}}{(s+5)^3}}$$

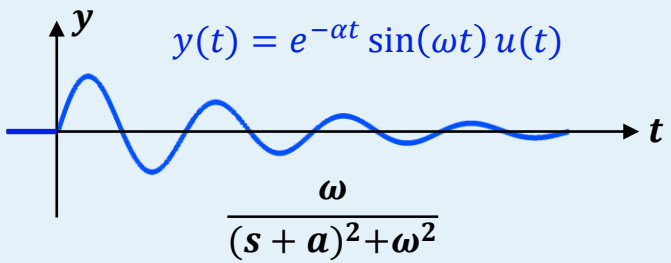
Stable & Unstable Responses

Considering single-sided responses of the form $y(t) = f(t)u(t)$, we call the response stable if $y(t)$ remains bounded for all $t \geq 0$, and unstable if it grows without bound.

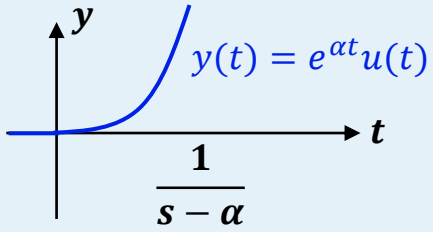
For example, the following functions are stable responses:

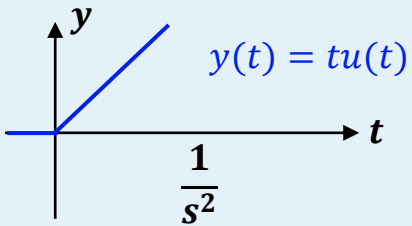


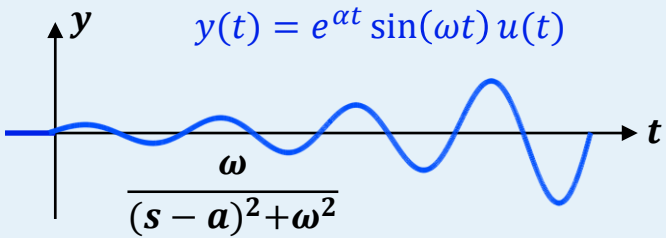




And the following functions are unstable responses:







Stable & Unstable Responses

We can determine stability from a Laplace transform $Y(s) = N(s)/D(s)$, after cancelling any common factors.

- Find the poles by solving: $D(s) = 0 \quad \Rightarrow \quad s = s_1, s_2, s_3, \ldots$
- The response is stable if every pole has a non-positive real part and every pole on the imaginary axis is simple.

$$y(t) = e^{-t}u(t) \quad \Rightarrow \quad Y(s) = \frac{1}{s+1} \quad \Rightarrow \quad s_1 = -1 \quad \Rightarrow \quad \mathrm{Re}\{s_1\} \leq 0 \quad \Rightarrow \text{ stable response}$$

$$y(t) = \sin(t)\,u(t) \quad \Rightarrow \quad Y(s) = \frac{1}{s^2+1} \quad \Rightarrow \quad s_{1,2} = \pm j \quad \Rightarrow \quad \begin{cases} \mathrm{Re}\{s_1\} = \mathrm{Re}\{s_2\} = 0 \\ s_1 \neq s_2 \text{ (simple roots)} \end{cases} \quad \Rightarrow \text{ stable response}$$

$$y(t) = e^{-t}\sin(t)\,u(t) \quad \Rightarrow \quad Y(s) = \frac{1}{(s+1)^2+1} \quad \Rightarrow \quad s_{1,2} = -1 \pm j \quad \Rightarrow \quad \begin{cases} \mathrm{Re}\{s_1\} = -1 < 0 \\ \mathrm{Re}\{s_2\} = -1 < 0 \end{cases} \quad \Rightarrow \text{ stable response}$$

- The response is unstable if at least one pole has a positive real part, or if any pole on the imaginary axis is repeated.

$$y(t) = e^{t}u(t) \quad \Rightarrow \quad Y(s) = \frac{1}{s-1} \quad \Rightarrow \quad s_1 = 1 \quad \Rightarrow \quad \mathrm{Re}\{s_1\} > 0 \quad \Rightarrow \text{ unstable response}$$

$$y(t) = tu(t) \quad \Rightarrow \quad Y(s) = \frac{1}{s^2} \quad \Rightarrow \quad s_{1,2} = 0 \quad \Rightarrow \quad \begin{cases} \mathrm{Re}\{s_1\} = \mathrm{Re}\{s_2\} = 0 \\ s_1 = s_2 \text{ (repeated root)} \end{cases} \quad \Rightarrow \text{ unstable response}$$

$$y(t) = e^{t}\sin(t)\,u(t) \quad \Rightarrow \quad Y(s) = \frac{1}{(s-1)^2+1} \quad \Rightarrow \quad s_{1,2} = 1 \pm j \quad \Rightarrow \quad \begin{cases} \mathrm{Re}\{s_1\} = +1 > 0 \\ \mathrm{Re}\{s_2\} = +1 > 0 \end{cases} \quad \Rightarrow \text{ unstable response}$$

Example

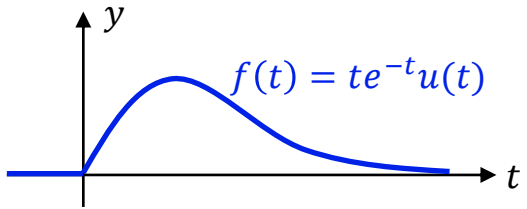
Find the Laplace Transform of the following functions, illustrate their poles in s-plane

(a) $f(t) = te^{-t}u(t)$ (b) $g(t) = t\cos(t)\,u(t)$

(a) $f(t) = te^{-t}u(t)$

Using the table of Laplace transforms:

$$\mathcal{L}(e^{-t}u(t)) = \frac{1}{s+1}$$

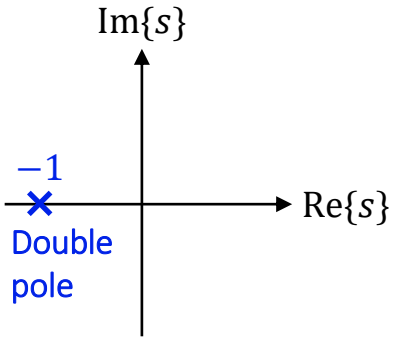


Applying $\mathcal{L}(ty(t)) = -\frac{\mathrm{d}}{\mathrm{d}s}Y(s)$:

$$F(s) = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{s+1}\right) = \frac{1}{(s+1)^2}$$

The poles satisfy: $(s+1)^2 = 0 \Rightarrow s_1 = s_2 = -1$

The real part of both roots is negative, so $f(t)$ is a stable response.



Example

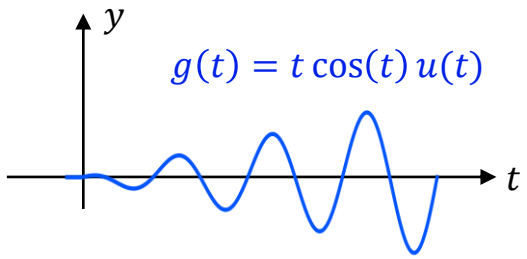
Find the Laplace Transform of the following functions, illustrate their poles in s-plane

(a) $f(t) = te^{-t}u(t)$ (b) $g(t) = t\cos(t)\,u(t)$

(b) $g(t) = t\cos(t)\,u(t)$

Using the table of Laplace transforms:

$$\mathcal{L}(\cos(t)\,u(t)) = \frac{s}{s^2+1}$$

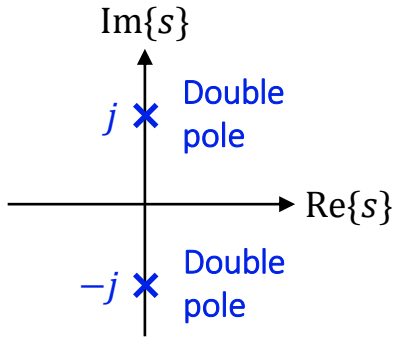


Applying $\mathcal{L}(ty(t)) = -\frac{\mathrm{d}}{\mathrm{d}s}Y(s)$:

$$G(s) = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{s}{s^2+1}\right) = \frac{s^2-1}{(s^2+1)^2}$$

The poles satisfy: $(s^2+1)^2 = 0 \Rightarrow s_{1,2} = j, s_{3,4} = -j$

There are repeated roots on the imaginary axis, so the response is unstable.

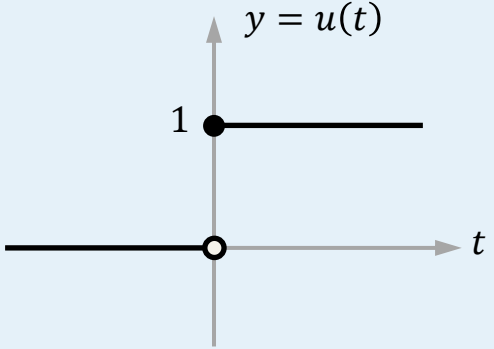


Summary

STEP FUNCTION

We define the step function as

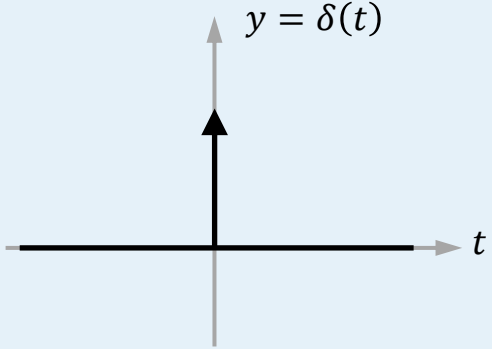
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$



DELTA FUNCTION

We define the delta function as

$$\delta(t) = \frac{\mathrm{d}}{\mathrm{d}t}u(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$



IMPORTANT PROPERTIES

1)

$$\int_{-\infty}^{\infty} \delta(t)\,\mathrm{d}t = 1$$

2)

$$\int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau = u(t)$$

3)

$$f(t) = \int_{-\infty}^{\infty} f(\tau)\delta(t-\tau)\,\mathrm{d}\tau$$

Summary

LAPLACE TRANSFORM

For a time-domain signal defined for $t > 0$

$$y(t) = f(t)u(t)$$

The Laplace transform of $y(t)$ is a function of a new parameter s, defined as

$$Y(s) = \mathcal{L}\big(y(t)\big) = \int_0^\infty y(t)e^{-st}\,\mathrm{d}t$$

$\mathcal{L}$ is a linear transformation

$$\mathcal{L}(ay_1 + by_2) = a\mathcal{L}(y_1) + b\mathcal{L}(y_2)$$

TABLE OF LAPLACE TRANSFORM

We can use this table to compute the Laplace Transform of functions:

$y(t)$	$Y(s)$	$y(t)$	$Y(s)$
$\delta(t)$	1	$e^{-at}f(t)u(t)$	$F(s+a)$
$u(t)$	$\frac{1}{s}$	$tf(t)u(t)$	$-F'(s)$
$\frac{1}{n!}t^n\,u(t)$	$\frac{1}{s^{n+1}}$	$f(t-t_0)u(t-t_0)$	$e^{-st_0}F(s)$
$\sin(\omega t)\,u(t)$	$\frac{\omega}{s^2+\omega^2}$	$f(t)*g(t)$	$F(s)G(s)$
$\cos(\omega t)\,u(t)$	$\frac{s}{s^2+\omega^2}$	$f'(t)u(t)$	$sF(s)-f(0)$
$e^{-at}\sin(\omega t)\,u(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$	$f''(t)u(t)$	$s^2F(s)-sf(0)-f'(0)$
$e^{-at}\cos(\omega t)\,u(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$	$\int_0^t f(\tau)u(\tau)\,\mathrm{d}\tau$	$\frac{1}{s}F(s)$

Tutorial Questions

NTU SINGAPORE

Tutorial 10 Laplace Transform Basics

1. Compute the Laplace Transforms *based on the definition of the Laplace Transform.*

   (a) $e^{-2t} \times u(t)$

   (b) $\sin 3t \times u(t)$

   (c) $t \times u(t)$

   (d) $(t-2) \times u(t-2)$

2. Sketch the function given by

   (a) $g(t) = 1$ for $0 \leq t \leq 2$
   $= 2$ for $2 \leq t \leq 3$
   $= 0$ elsewhere

   Write $g(t)$ in terms of (delayed) step functions.

   (b) What is the Laplace Transform of $g(t)$ in part (a)?

3. With the aid of basic properties and basic Laplace Transforms, find the Laplace Transforms of

   (a) $\sin(t-2)u(t-2)$

   (b) $e^{-(t-2)}\sin(t-2)u(t-2)$

   (c) $e^{-t}\sin(t-2)u(t-2)$

   (d) $\cos^2(t)u(t)$